An Automated Early Warning System for Learner Sentiment using Robotic Process Automation and Knowledge-Based Systems

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Abstract. Higher Education Institutions (HEIs) increasingly require efficient systems to detect and address learners' negative emotions expressed on social networks. Manual monitoring and traditional rule-based methods are inadequate for processing large-scale unstructured data, often leading to delays and low accuracy. To overcome these challenges, this study proposes an automated sentiment analysis framework that integrates Robotic Process Automation (RPA) and Knowledge-Based Systems (KBS). The RPA module, implemented in Python, automatically collects legitimate data from HEIs' social networks, extracts content and metadata, and delivers alerts to relevant departments. The KBS module then classifies learner feedback using PhoBERT embeddings and cosine similarity, enabling precise mapping of negative sentiment to the appropriate units. A pilot implementation at a faculty in a university in Vietnam resulted in a classification accuracy of around 80%, indicating that the integrated approach enhances operational efficiency, reduces manual workload, and ensures the provision of reliable and timely early warnings. Findings suggest that the proposed learner-centered early warning system is effective in enhancing learner satisfaction and engagement. In addition, with the advantage of being easily scalable and replicable, the system is suitable for application across faculties, HEIs, and diverse application contexts.

Keywords: Robotic Process Automation, Knowledge-Based Systems, Sentiment Analysis, Early Warning System, Higher Education.

1 Introduction

Within HEIs, individual departments face the persistent challenge of continuously monitoring and understanding the views of the learners they serve [1, 2, 7]. Regularly reviewing and analyzing learner feedback on social networks requires a significant amount of human effort, and the repetitive nature of this task makes it both resource-intensive and error-prone [2, 5].

This situation highlights the necessity of developing a systematic approach that reduces manual workload while ensuring timely and accurate insights. Specifically, three key needs emerge. First, **automated data collection**, the system must be capable of continuously and lawfully gathering learner-related data from communication channels, such as official websites or social networks. Second, **evaluation and classification**, effective analysis of the collected feedback requires categorization not only by sentiment (positive, negative, neutral), where the integration of Long Short-Term Memory (LSTM) and Convolutional Neural Network (CNN) enables the modeling of sequential dependencies and improves sentiment classification performance [2], but also helps to identify its relationship to faculties, departments or stages of the training process. Third, **timely alerts**, the system should provide immediate warnings to department leaders, enabling them to detect negative learner perspectives early and respond with appropriate interventions [1, 2].

These requirements create the foundation for an automated warning system that integrates RPA, sentiment analysis models, and KBS to enhance transparency, efficiency, responsiveness in managing learner feedback content. The study is guided by two core research questions as follows:

- **RQ1:** How can HEIs automatically detect learners' negative sentiments at an early stage before they escalate into bigger issues?
- **RQ2:** How can an automated early warning system be effectively developed using RPA and KBS to support timely interventions?

With **RQ1**, the solution lies in designing an automated system that continuously monitors learner-generated content across official digital platforms. Rather than relying on manual reports or isolated complaints, the system automatically scans new posts and comments, analyzes their sentiment, flags negative feedback, and routes it to the appropriate departments. This enables institutions to detect concerns at an early stage and intervene before they escalate, functioning as an early warning mechanism that provides stakeholders with a timely and reliable overview of learner sentiment. Meanwhile, **RQ2** emphasizes how such a system can be developed in a way that ensures efficiency, accuracy, and transparency. The proposed framework leverages RPA to lawfully collect and structure data from institutional platforms (e.g., websites, fanpages, and groups), while KBS and sentiment analysis models handle classification and routing. This integration guarantees both transparency and legality of data collection, while automated workflows reduce manual workload, maintain fairness in interpreting learner feedback, and strengthen institutional trust.

Addressing these two research questions leads to the development of a scalable and accurate early warning system for learner sentiment, tailored to the needs of HEIs. Beyond higher education, the framework also has potential for broader application in domains such as customer service, business management, and e-government. In doing so, it contributes not only practical solutions but also methodological insights into the integration of RPA, KBS, and sentiment analysis for real-time unstructured data exploitation.

2 Methodology

2.1 Automated data collection using RPA

RPA is a powerful solution for automating repetitive tasks in sentiment analysis [2, 9]. While traditional RPA tools offer low-code environments and GUI automation, in this study we implemented a Python-based pipeline with two stages: **RPA data crawling** and **RPA warning notification**. In the first stage, the system automatically scans posts and comments on social networking platforms (such as fanpages, groups, etc.) managed by the faculty through the use of legal APIs provided by social networks. Using these APIs helps extract content (author, timestamp, metadata, etc.) completely and accurately. In addition, RPA also helps set the frequency of continuous data collection, ensuring that the data is always complete and synchronized. In the second stage, posts and comments labeled as negative by sentiment analysis are used to generate alert emails with key details, which are then sent to the relevant department via SMTP [2, 10]. Both stages run unattended on a fixed schedule, replacing manual work with programmatic scripts and demonstrating the three core traits of RPA: task automation, autonomous operation, and scheduled execution.

Through the use of RPA, it helps improve the performance of sentiment analysis systems [2]. Automated data collection is the basis for performing real-time sentiment trend analysis. With the analysis results, automatically identifying and sending the results to the relevant departments in a timely manner via email allows for timely intervention, helping HEIs increase engagement with learners.

2.2 Using KBS to segment departments

After automatically collecting more than 1,500 posts and 7,500 comments shared by learners on social networks at a faculty in a Vietnamese university through RPA, KBS was used to segment the data to determine which units in the university those data were related to.

Methods for utilizing KBS: The system retrieves and manages keywords related to each unit/department in Vietnamese university directly from the database, thus creating a dynamic knowledge base that supports real-time updates. The collected data are organized into a dictionary and stored in memory at initialization, while also being represented in JSON format to facilitate formatting, transmission, and processing. This approach ensures fast query performance and efficient resource utilization. Any modification to the keywords or departmental structure in the database automatically triggers a synchronization process, thereby maintaining consistency between the stored data and the internal processing workflow. Direct integration with the database simplifies the system while preserving flexibility, particularly in environments where data are frequently updated across multiple tables and sources within the same database management system.

Application of Cosine Similarity with Sentiment Analysis Models The system implements an automatic classification method, in which the input

text is processed by a sentiment analysis model. Then, it is compared with keywords related to faculties/departments using the cosine similarity method. Thereby, the negative content of the learner can be quickly identified as related to which faculties/departments in the university.

Cosine similarity is computed as in Eq. (1)

$$\text{Cosine Similarity}(A, B) = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \sqrt{\sum_{i=1}^{n} B_i^2}} \tag{1}$$

with

- $\sum_{i=1}^{n} A_i B_i$: dot product of A and B,
- $\sqrt{\sum_{i=1}^{n} A_i^2}$ and $\sqrt{\sum_{i=1}^{n} B_i^2}$: Euclidean norms of A and B.

To enable automatic classification, the system integrates PhoBERT—a deep learning model pretrained on large-scale Vietnamese text corpora [2, 8]—in combination with the cosine similarity approach [4]. By examining similarity scores ranging from 0 to 1, the system can accurately identify the departments or units most relevant to the given input.

Cosine similarity offers several notable advantages in recommendation systems and natural language processing tasks [4]. It measures the degree of similarity between sentences based on the angle between their vector representations, ensuring robustness against variations in sentence length. When applied to sentence embeddings generated by PhoBERT, cosine similarity captures the semantic characteristics of sentences without worrying about differences in sentence length [11]. Furthermore, when combined with extraction techniques from transformer models, cosine similarity not only improves the performance of embedding models but also makes the system more flexible [6].

3 Results

3.1 Proposed workflow for data collection, analysis, and warning notification using RPA and KBS

After identifying the problem and considering an appropriate implementation solution to collect and categorize learners' opinion data related to departments within a Vietnamese university, a proposed process for data collection, classification, and alert notification is presented in the Figure 1.

In this process, the process starts by automatically collecting data from the official fanpage of the faculty through APIs provided by social networks. These contents are then passed to a sentiment analysis module, where a deep learning architecture enhanced with an LSTM and CNN layer is used to capture the sequential patterns of Vietnamese sentences and improve the recognition of negative sentiment [2]. Only the posts/comments flagged as negative continue to the next stage, where they are transformed into embeddings and matched against department-specific keyword embeddings maintained in a dynamically updated knowledge base. Cosine similarity is applied to measure closeness, and a

threshold-based rule identifies the most relevant faculties/departments. Finally, the system sends automated email alerts to notify the learner's concerns to the relevant faculties/departments, which is the basis for timely responses. This comprehensive workflow integrates automated data collection, advanced sentiment analysis, and embedding-based classification to ensure proactive handling of learner feedback.

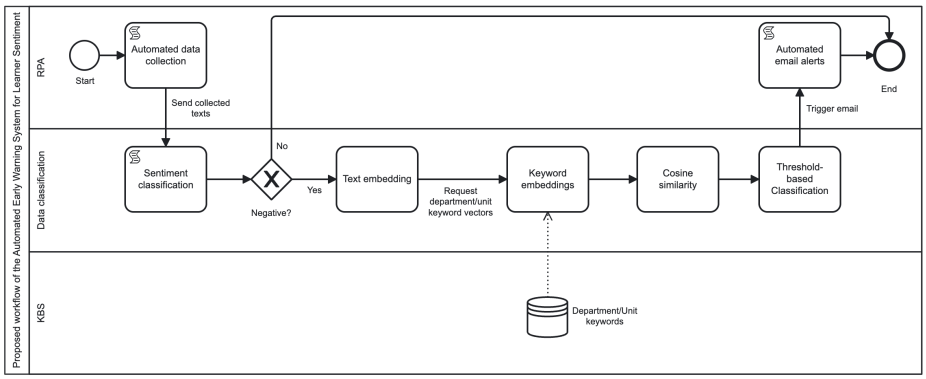

Fig. 1: Proposed workflow of the Automated Early Warning System for Learner Sentiment

3.2 Applying Python-based RPA to automatically collect data and send warning emails

In order to systematically monitor learner feedback within online communities, we developed a framework that automates the collection, processing, and legitimization of information from social media platforms managed by HEIs. The system is designed to ensure greater efficiency and reliability, while maintaining methodological rigor in data extraction, sentiment analysis, and targeted notification.

RPA data crawling: This phase begins with a list of official social media page URLs (fanpage, group, etc.) managed by a department or unit within the university. The headless browser then loads the content via the APIs provided by the respective social media platform. From each post, essential details—such as the content and comments, author, timestamp, likes, shares, and attached links—are extracted and stored in a structured format. To ensure legal and ethical compliance, the data collection process follows the Family Educational Rights and Privacy Act (FERPA) and General Data Protection Regulation (GDPR) standards. Specifically, any personally identifiable information (PII) is either removed or anonymized before storage and further analysis. Techniques such as de-identification (removing usernames, IDs, or personal attributes) and pseudonymization (replacing identifiers with system-generated codes) are applied to protect the privacy of individuals while preserving the analytical value of the dataset. As a result, the pipeline not only provides a reliable and transparent method of data gathering but also ensures fairness, legality, and respect

for user privacy. The process can be executed through a headless browser (running in the background, as illustrated in Figure 2) or via a standard browser for real-time, visual monitoring.

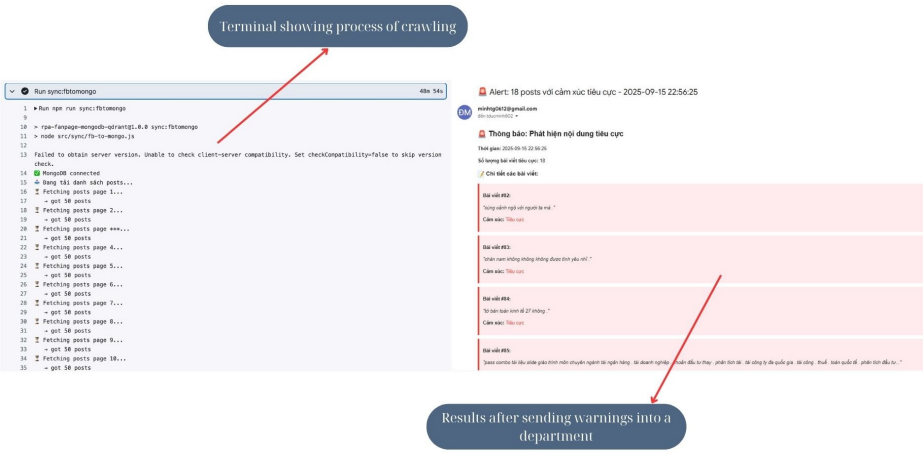

Fig. 2: Process of data crawling and warning notification using Python-based Robotic Process Automation

RPA warning notification: At the final stage, the system will select all posts that are labeled negative and tag the responsible department. For each review, an email will be generated, including the post URL, content excerpt, sentiment label, author and department, along with a standardized subject line. Using the Python email library, the system will automatically send alerts to the relevant department/unit address, applying batch processing or throttling to avoid spam flags. All alerts are logged with timestamps, and errors will trigger retries or error reporting, ensuring reliability and consistency in the notification process. The results can be seen on the right side of Figure 2.

In summary, the Python-based RPA system we developed has ensured automation, efficiency, speed and reliability as the work of a human analyst in monitoring social networks.

3.3 Evaluation of the retrieval and processing capabilities of KBS

KBS is implemented with the ability to directly retrieve data from the database, perform automatic synchronization, and store information in memory to optimize processing speed. To evaluate its performance in real-time query scenarios, experiments were conducted with multiple metrics, including initialization time, query latency, throughput, success rate, and cache utilization. The detailed results are presented in Table 1.

The experimental results demonstrate that KBS operates reliably with a dataset comprising 11 faculties/departments and 1,100 keywords, using 1,688 MB of memory and a cache of 1,445 items (Table 1). Although the initialization time remains relatively high (9.224 s, including 3.213 s for keyword loading

and 1.257 s for embedding computation), the subsequent intermediate processes are highly efficient: classification (0.119 s), embedding creation (0.120 s), and especially similarity computation (0.003 s). Query response times range from 0.002–0.308 s, with a throughput of 5.8 queries/s, indicating acceptable concurrent processing capability. The system attained an overall success rate of 80.6% across 1,486 queries, although the cache hit rate remained relatively low at 1.7%, underscoring the need for refinement of caching strategies. Nevertheless, the system demonstrates sufficient robustness for practical deployment in environments characterized by frequently changing data, with potential gains in performance and scalability achievable through enhanced initialization and caching techniques.

Metric	**Value**
System Information	
Faculties/Departments	11
Keywords	1,100
Cache size	1,445 items
Memory usage	1,688.0 MB
Performance Metrics	
Initialization time	9.224 s
Load Keywords	3.213 s
Compute Embeddings	1.257 s
Average classification	0.119 s
Average embedding creation	0.120 s
Average similarity computation	0.003 s
Fastest query	0.002 s
Slowest query	0.308 s
Throughput	5.8 queries/s
Query Statistics	
Total queries	1,486
Successful queries	1,197
Failed queries	289
Success rate	80.6%
Cache hit rate	1.7%

Table 1: Query Performance of the Knowledge-Based System

3.4 Using the PhoBERT language model for department classification

To assess the performance of the proposed method, over 9000 learner posts and comments were automatically collected, among which 150 manually labeled samples were used for training and testing to analyze the Accuracy, Recall, Precision, and F1-Score of the PhoBERT model at different thresholds. The dataset provides strong authenticity by reflecting real learner contexts and issues, while cosine similarity was utilized to benchmark computed results against manual classifications, ensuring robust assessment of classification effectiveness of PhoBERT.

The experimental results presented in Table 2 demonstrate substantial variation in the performance of PhoBERT across different classification thresholds.

At lower thresholds (30%–45%), Recall consistently achieves 1.000, confirming the model's ability to capture all target classes. However, Precision remains very low (0.202–0.227), which limits both Accuracy (0.20–0.31) and the F1-Score (0.33–0.37). At medium thresholds (50%–60%), the metrics become more balanced. Notably, the 60% threshold delivers the best trade-off, with Accuracy of 0.859, Recall of 0.950, Precision of 0.594, and an F1-Score of 0.731, making it the optimal setting for practical use. In contrast, higher thresholds (65%–70%) increase Precision (0.667–0.750) but sharply reduce Recall (0.600–0.300), resulting in a lower F1-Score. These findings indicate that overly strict thresholds cause many relevant cases to be overlooked. Overall, the evidence highlights the 60% threshold as the most suitable configuration, ensuring both high accuracy and robust detection. Accordingly, PhoBERT can be considered an effective choice for department-level classification of learner-generated content.

Threshold (%)	**Accuracy**	**Recall**	**Precision**	**F1-Score**
30	0.202	1.000	0.202	0.336
35	0.202	1.000	0.202	0.336
40	0.253	1.000	0.213	0.351
45	0.313	1.000	0.227	0.370
50	0.465	1.000	0.274	0.430
55	0.677	1.000	0.436	0.607
60	0.859	0.950	0.594	0.731
65	0.859	0.600	0.667	0.632
70	0.838	0.300	0.750	0.429

Table 2: PhoBERT model performance at different classification thresholds

4 Discussion

The Python-based RPA component effectively automated the collection and monitoring of learner feedback from university-managed social networks. It enabled real-time crawling, metadata extraction, and the delivery of structured alerts to relevant departments and units, thereby significantly reducing manual workload. Nevertheless, compared with traditional RPA platforms, the Python implementation is less user-friendly for non-technical staff, which may limit adoption in institutions without strong IT support.

The integration of KBS with PhoBERT embeddings and cosine similarity proved effective in classifying feedback by department, achieving high accuracy and balanced performance at the 60% similarity threshold. However, several limitations remain: the cache hit rate was low, embedding initialization required considerable time, and the system relied heavily on manually curated keyword lists. These factors constrain efficiency and adaptability, highlighting the need for automated keyword expansion, optimization of caching mechanisms, and faster initialization strategies.

Taken together, the combination of RPA and KBS creates a balanced framework that integrates large-scale automation with contextual intelligence. RPA ensures scalable and continuous data collection, while embedding-based classification enhances accuracy in routing feedback to the appropriate stakeholders.

By complementing each other's strengths, the two components produce a robust sentiment analysis and early-warning system that not only improves operational efficiency but also provides a reliable foundation for broader deployment.

Beyond technical contributions, the proposed learner sentiment early warning system has important social implications. If deployed across an entire university, it could significantly enhance two-way communication between learners and institutional stakeholders. By enabling timely detection of concerns and structured reporting, administrators and faculty members can respond proactively, fostering transparency, trust, and improved learner satisfaction. Such responsiveness is especially critical in higher education, where engagement and retention are closely tied to the perceived quality of institutional support.

Furthermore, the architecture is inherently generalizable to other domains. In customer service, it could be employed to monitor and address dissatisfaction expressed on digital platforms. In business management, it offers the ability to capture employee sentiment and identify organizational challenges at an early stage. In e-government, the framework could support real-time analysis of citizen feedback from online portals and social media, thereby facilitating more responsive governance and improving service delivery. These cross-sectoral applications demonstrate the scalability and versatility of the proposed approach.

5 Conclusion and Future work

This study introduced a practical early-warning system that integrates RPA and KBS for monitoring learner sentiment in HEIs. By automatically collecting learner-generated posts and comments, and applying PhoBERT embeddings with cosine similarity for classification, the framework achieved around 80% accuracy in detecting negative sentiment. A pilot deployment at a Vietnamese faculty demonstrated its effectiveness in reducing manual monitoring effort and providing timely alerts to relevant departments.

Despite these promising results, several limitations remain. The system still encounters performance bottlenecks such as slow initialization, suboptimal caching, and occasional query failures under large-scale workloads. Its adaptability is constrained by reliance on manually curated keyword sets and a relatively small labeled dataset, which reduce responsiveness to evolving language patterns and emerging learner concerns.

Future improvements can address these challenges along three directions:

- **Model enhancement**: fine-tuning PhoBERT on larger datasets and integrating topic modeling or summarization techniques to generate concise and actionable reports.
- **System scalability**: optimizing caching, automating keyword expansion, and improving initialization speed for more efficient large-scale deployment.
- **User-centered features**: developing dashboards, chatbot interfaces [3], and predictive modules to provide real-time insights and identify rising negative sentiment before escalation.

In conclusion, the integrated RPA–KBS framework demonstrates significant potential as a scalable and effective solution for learner sentiment monitoring. With further refinements, it can evolve into a robust decision-support tool that enhances transparency, responsiveness, and engagement not only in HEIs but also across other sectors where timely analysis of user feedback is critical.

References

1. Chu, V.H., Pham, X.L.: Application of Robotic Process Automation in collection of students satisfaction at University. In: 2023 14th IIAI International Congress on Advanced Applied Informatics (IIAI-AAI). pp. 604–609 (2023). https://doi.org/10.1109/IIAI-AAI59060.2023.00120
2. Chu, V.H., Pham, X.L., Nguyen, T.S.: Combining RPA and AI to recognize learners' emotions in Cyberspace: Experiment at some Vietnamese universities. In: 2025 7th International Conference on Computer Science and Technologies in Education (CSTE). pp. 464–470 (2025). https://doi.org/10.1109/CSTE64638.2025.11092155
3. Clarizia, F., Colace, F., Lombardi, M., Pascale, F., Santaniello, D.: Chatbot: An education support system for student. In: International Symposium on Cyberspace Safety and Security. pp. 291–302. Springer International Publishing (2018). https://doi.org/10.1007/978-3-030-01689-0_23
4. Gunawan, D., Sembiring, C.A., Budiman, M.A.: The Implementation of Cosine Similarity to Calculate Text Relevance Between Two Documents. In: Journal of Physics: Conference Series. vol. 978, p. 012120. IOP Publishing (March 2018). https://doi.org/10.1088/1742-6596/978/1/012120
5. Henderson, M., Ryan, T., Phillips, M.: The challenges of feedback in higher education. Assessment & Evaluation in Higher Education **44**(8), 1237–1252 (2019). https://doi.org/10.1080/02602938.2019.1599815
6. Laskar, M.T.R., Huang, X., Hoque, E.: Contextualized embeddings based transformer encoder for sentence similarity modeling in answer selection task. In: Proceedings of the Twelfth Language Resources and Evaluation Conference (LREC). pp. 5505–5514 (May 2020), https://aclanthology.org/2020.lrec-1.676
7. Mukit, M.M.H., Chu, V.H., Pham, X.L., Kakatum, S.R., Srivastava, N.: Design and Evaluation of a Hybrid RPA-AI Framework for Intelligent Transcript Processing in Higher Education. International Journal of Computational and Experimental Science and Engineering **11**(3) (Sep 2025). https://doi.org/10.22399/ijcesen.3866
8. Pham Duy, A., Le Thanh, H.: A Question-Answering System for Vietnamese Public Administrative Services. In: Proceedings of the 12th International Symposium on Information and Communication Technology. pp. 85–92 (December 2023). https://doi.org/10.1145/3628797.3628965
9. Sharma, V., et al.: Robotic Process Automation Applications in Data Management. In: AI-Centric Modeling and Analytics, pp. 238–259. CRC Press (2023)
10. Sudana, I.M., Qudus, N., Prasetyo, S.E.: Implementation of PHPMailer with SMTP protocol in the development of web-based e-learning prototype. In: Journal of Physics: Conference Series. vol. 1321. IOP Publishing (2019). https://doi.org/10.1088/1742-6596/1321/3/032027
11. Thanh, T.D., Nguyen, T.M., Nguyen, T.B., Nguyen, H.T., Nguyen, H.L., Tran, M.V., Nguyen, H.T.: Graph-based and generative approaches to Multi-Document summarization. Journal of Computer Science and Cybernetics **40**(3), 203–217 (2024). https://doi.org/10.15625/1813-9663/18353